

SSC CGL

DAY 4 COMPREHENSIVE NOTES

SUBJECT 1: Arithmetic — Profit, Loss & Discount

SUBJECT 2: Classification — Odd One Out

SUBJECT 3: Grammar — Active & Passive Voice

SUBJECT 4: Static GK — Indian Polity

Weightage: 2–3 (Math) | 2–4 (Classification) | Moderate–High (Grammar) | 8–10 (Polity Tier 1)

Compiled with formulae, shortcuts, solved PYQs, and quick revision tables

SUBJECT 1: ARITHMETIC — PROFIT, LOSS & DISCOUNT

PART 1: Basic Concepts & Formulae

Cost Price (CP): Price at which article is purchased.

- $CP = SP - \text{Profit (when profit)}$ | $CP = SP + \text{Loss (when loss)}$

Selling Price (SP): Price at which article is sold.

- $SP = CP + \text{Profit}$ | $SP = CP - \text{Loss}$

Profit (SP > CP): Profit = $SP - CP$ | Profit% = $(\text{Profit}/CP) \times 100$

Loss (CP > SP): Loss = $CP - SP$ | Loss% = $(\text{Loss}/CP) \times 100$

PART 2: Quick Memory Table

Scenario	Formula
Profit %	$(SP - CP)/CP \times 100$
Loss %	$(CP - SP)/CP \times 100$
SP when Profit	$CP \times (100 + P)/100$
SP when Loss	$CP \times (100 - L)/100$
CP when Profit	$SP \times 100/(100 + P)$
CP when Loss	$SP \times 100/(100 - L)$

PART 3: Shortcut Tricks

Trick 1 – CP of x = SP of y: Profit% = $[(x - y)/y] \times 100$

Trick 2 – Successive P/L: Net% = $a + b + (ab/100)$

Trick 3 – False Weight: Profit% = $[(\text{True} - \text{False})/\text{False}] \times 100$

Trick 4 – CP of 1 = SP of 1: $CP = SP \times 100/(100 + P\%)$

PART 4: Discount Concepts

Discount is always on MP; Profit/Loss is always on CP.

Concept	Formula
SP after Discount	$MP \times (100 - D)/100$
MP given SP & D%	$SP \times 100/(100 - D)$
Profit% after Discount	$(SP - CP)/CP \times 100$

PART 5: Advanced Concepts

Equivalent Discount (a%, b%): $a + b - (ab/100)$

Net% when MP x% above CP and Discount y%: $x - y - (xy/100)$

PART 6: Solved PYQs

Type	Question	Solution
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Basic	CP=■240, Profit=12%	SP=268.80
Basic	SP=■720, Loss=10%	CP=800
CPx=SPy	CP of 15 = SP of 12	Profit=25%
Successive	20% then 10% profit	Total=32%
False Wt	10% profit, 20% less weight	Profit=37.5%
Discount	MP=500, 20% off	SP=400
MP-CP	MP 40% above CP, Disc 20%	Net Profit=12%
Equiv Disc	Successive 20% & 10%	28%

PART 7: Golden Rules

- Always calculate profit/loss on CP
- Always calculate discount on MP
- If in doubt, take CP = ■100
- For false weight, base is false weight
- In successive changes, use $\pm a \pm b \pm (ab/100)$

SUBJECT 2: CLASSIFICATION — ODD ONE OUT

Golden Rule: Three have a common property, one doesn't.

PART 1: Word Classification

Type	Options	Odd One	Reason
GK	Mercury,Venus,Earth,Mars,Jupiter	Jupiter	Others terrestrial; Jupiter gaseous
Gender	Cow,Mare,Bull,Doe,Hen	Bull	Others are female
Profession	Doctor,Engineer,Tailor,Teacher	Tailor	Others need degree
Part-Whole	Wheel,Door,Window,Steering	Steering	Inside vs outside
Weapon	Knife,Sword,Gun,Dagger	Gun	Others sharp-edged
Synonym	Happy,Joyful,Sad,Cheerful,Glad	Sad	Others = happiness
Vowel	Apple,Orange,Elephant,Ink,Umbrella	Orange	Consonant start

PART 2: Number Classification

Type	Options	Odd One	Reason
Prime	2,3,5,9,7	9	Others prime
Square	4,9,16,20,25	20	Not a perfect square
Cube	8,27,64,81,125	81	81 is square, not cube
Even	2,4,6,9,10	9	Others even
Divisible	12,18,24,25,30	25	Not divisible by 6

PART 3: Figure Classification

Type	Logic
Shape	Quadrilaterals vs Triangle
Direction	Missing arrow in sequence
Rotation	Non-multiple-of-90° rotation
Open/Closed	Line vs Polygon
Symmetry	Shapes lacking line symmetry

PART 4: Practice Set

- Car,Bus,Bicycle,Truck → **Bicycle** (others run on fuel)
- Gold,Silver,Wood,Copper → **Wood** (others metals)
- 27,64,100,125 → **100** (others perfect cubes)
- 11,13,15,17 → **15** (others prime)
- 2-3,5-7,9-11,11-13 → **9-11** (9 isn't prime)

SUBJECT 3: GRAMMAR — ACTIVE & PASSIVE VOICE

PART 1: Tense-wise Conversion

Tense	Active	Passive
Present Indefinite	Sub+V1(s/es)+Obj	Obj+is/am/are+V3
Present Continuous	Sub+is/am/are+V-ing+Obj	Obj+is/am/are+being+V3
Present Perfect	Sub+has/have+V3+Obj	Obj+has/have+been+V3
Past Indefinite	Sub+V2+Obj	Obj+was/were+V3
Past Continuous	Sub+was/were+V-ing+Obj	Obj+was/were+being+V3
Past Perfect	Sub+had+V3+Obj	Obj+had+been+V3
Future Indefinite	Sub+will/shall+V1+Obj	Obj+will/shall+be+V3
Future Perfect	Sub+will/shall+have+V3+Obj	Obj+will/shall+have+been+V3

PART 2: Special Rules

Modals: Obj + Modal + be + V3

Imperatives: Let + Obj + be + V3 (Order) | Obj + should be + V3 (Advice) | You are requested to + V (Request)

Infinitives: Sub + Verb + to be + V3 + Obj

Gerunds: Sub + Verb + being + V3 + Obj

Double Objects: A book was given to him / He was given a book

Preposition: Keep preposition after V3 — 'looked at', 'insisted on'

Intransitive: Sleep, Run, Go, Come, Die — no passive (unless with preposition)

PART 3: Important V3 Forms

Write→Written	Eat→Eaten	Drink→Drunk
Take→Taken	Give→Given	See→Seen
Know→Known	Speak→Spoken	Choose→Chosen
Drive→Driven	Break→Broken	Forget→Forgotten

PART 4: Golden Rules

- Subject of Active → Object of Passive; Object → Subject
- Add auxiliary (be/being/been) + V3
- Use 'by' before agent; omit if unknown/obvious
- In interrogatives, keep question word at beginning
- If verb has preposition, keep it right after V3

SUBJECT 4: STATIC GK — INDIAN POLITY

PART 1: Constitution Overview

Adopted: November 26, 1949 | **Enforced:** January 26, 1950

Length: Longest written constitution | **Parts:** 25 | **Schedules:** 12 | **Articles:** 448 (originally 395)

Drafting Committee Chairman: Dr. B.R. Ambedkar

Permanent Chairman, CA: Dr. Rajendra Prasad

PART 2: Sources of the Constitution

Borrowed From	Features
Govt of India Act 1935	Federal scheme, Governor, Emergency powers, Federal judiciary
Britain (UK)	Parliamentary system, Rule of Law, Single citizenship, Law-making
USA	Fundamental Rights, Judicial Review, Preamble, Independence of Judiciary
Canada	Federation with strong Centre, Residuary powers to Centre
Ireland	DPSP, Presidential election, Rajya Sabha nominations
Germany	Suspension of FRs during Emergency
Australia	Concurrent List
South Africa	Amendment with 2/3rd majority, RS proportional representation

PART 3: Important Parts & Articles

Part	Articles	Subject
Part I	1–4	Union and its Territory
Part II	5–11	Citizenship
Part III	12–35	Fundamental Rights
Part IV	36–51	Directive Principles of State Policy
Part IVA	51A	Fundamental Duties
Part V	52–151	Union Government
Part VI	152–237	State Government
Part IX	243–243O	Panchayats
Part IXA	243P–243ZG	Municipalities
Part XI	245–263	Centre-State Relations
Part XV	324–329	Elections & EC
Part XVIII	352–360	Emergency Provisions
Part XX	368	Amendment of Constitution

PART 4: Important Schedules

Schedule	Deals With
1st	States & Union Territories
8th	Official Languages (22 languages)
10th	Anti-Defection Law
12th	Municipalities

PART 5: Key Amendments

Amendment	Year	Key Provisions
42nd	1976	Mini Constitution — Added Socialist, Secular, Integrity to Preamble; Added Fundamental Duties (51A)
44th	1978	Restored FRs; Right to Property removed as FR (became legal right under Art. 300A)
73rd	1992	Panchayati Raj Institutions (Part IX)
74th	1992	Municipalities (Part IXA)
86th	2002	Right to Education (Art. 21A) — 6–14 years
101st	2016	Goods & Services Tax (GST)

PART 6: The Preamble

Proposed by: Jawaharlal Nehru (Objectives Resolution)

First Amendment: 42nd Amendment (1976) — added Socialist, Secular, Integrity

Called 'Keynote to Constitution': Ernest Barker

Called 'Horoscope of Republic': K.M. Munshi

Source of Preamble idea: Constitution of USA

Keywords in Preamble:

Sovereign — Independent; free from external control

Socialist — Commitment to social justice & welfare (42nd Amendment, 1976)

Secular — No state religion; equal treatment of all religions (42nd Amendment, 1976)

Democratic — Government of the people, by the people, for the people

Republic — Head of State is elected (not hereditary)

PART 7: Fundamental Rights — Part III (Articles 12–35)

Total Rights Originally: 7 | **Currently:** 6 (Right to Property removed by 44th Amendment)

Called 'Heart & Soul of Constitution' by Dr. B.R. Ambedkar: Article 32

1. Right to Equality (Articles 14–18)

Art. 14: Equality before law & equal protection of laws

Art. 15: Prohibition of discrimination (religion, race, caste, sex, place of birth)

Art. 16: Equality of opportunity in public employment

Art. 17: Abolition of Untouchability (enforceable against private individuals)

Art. 18: Abolition of titles (except military/academic)

2. Right to Freedom (Articles 19–22)

Art. 19: Six Freedoms — speech, assembly, association, movement, residence, profession

Art. 20: Protection in respect of conviction (ex-post facto, double jeopardy, self-incrimination)

Art. 21: Right to Life & Personal Liberty (most expansive; includes livelihood, privacy, education)

Art. 21A: Right to Education (6–14 years) — 86th Amendment, 2002

Art. 22: Protection against arrest & detention

3. Right Against Exploitation (Articles 23–24)

Art. 23: Prohibition of traffic in human beings & forced labour (begar)

Art. 24: Prohibition of child labour below 14 years in factories/mines

4. Right to Freedom of Religion (Articles 25–28)

Art. 25: Freedom of conscience & profession, practice, propagation of religion

Art. 26: Freedom to manage religious affairs

Art. 27: Freedom from taxation for promotion of any religion

Art. 28: Freedom to attend religious instruction in educational institutions

5. Cultural & Educational Rights (Articles 29–30)

Art. 29: Protection of interests of minorities (language, script, culture)

Art. 30: Right of minorities to establish & administer educational institutions

6. Right to Constitutional Remedies (Articles 32–35)

Art. 32: Right to move Supreme Court for enforcement of FRs — 'Heart & Soul'

Art. 33: Power to restrict FRs for Armed Forces

PART 8: Writs Under Article 32 & 226

Writ	Meaning
Habeas Corpus	Produce the body — against unlawful detention
Mandamus	We command — compel public authority to perform duty
Prohibition	To forbid — stop lower court from exceeding jurisdiction
Certiorari	To be certified — quash order of lower court
Quo Warranto	By what authority — question right to hold public office

PART 9: Fundamental Duties — Part IVA (Article 51A)

Added by: 42nd Amendment, 1976 | **Inspired by:** USSR Constitution

Originally: 10 Duties | **Now:** 11 Duties (86th Amendment added duty for parents to provide education to children 6–14)

- Abide by Constitution & respect national flag/anthem
- Cherish ideals of freedom struggle
- Protect sovereignty & integrity of India
- Defend the country
- Promote harmony & spirit of common brotherhood
- Preserve rich heritage & composite culture
- Protect & improve natural environment
- Develop scientific temper & humanism
- Safeguard public property
- Strive towards excellence

PART 10: Quick Revision Table

Topic	Key Point
Constitution Adoption	26 Nov 1949
Constitution Enforcement	26 Jan 1950
Drafting Committee Chairman	Dr. B.R. Ambedkar
Permanent Chairman of CA	Dr. Rajendra Prasad
Preamble Borrowed From	USA Constitution
DPSP Borrowed From	Ireland Constitution
Fundamental Duties Borrowed From	USSR Constitution

Preamble Amendment	42nd Amendment, 1976
Heart & Soul	Article 32
FR Removed	Right to Property (44th Amendment, 1978)
Right to Education	Article 21A (86th Amendment, 2002)
Fundamental Duties	Article 51A (42nd Amendment, 1976)
Emergency Provisions	Articles 352–360
Constitutional Amendment	Article 368

PART 11: Golden Rules for Polity

- Focus on key articles: Art. 14, 19, 21, 21A, 32, 356, 368
- Know the amendments: 42nd (Mini Constitution), 44th (Restored FRs), 73rd & 74th (Local Self-Govt)
- Don't confuse sources: Preamble→USA | DPSP→Ireland | FD→USSR
- Remember: Ambedkar (Drafting) | Nehru (Objectives Resolution) | Rajendra Prasad (Permanent Chairman)

PART 12: Exam Day Tips

- Read carefully — look for 'not,' 'except,' 'originally'
- Know exact article numbers for key provisions
- Remember amendment numbers for key changes
- Don't spend more than 30 sec/question
- Eliminate obvious wrong options first

Final Mantra: 'Constitution is the framework, Preamble is the soul, Fundamental Rights are the guarantees — know these three, and you've mastered the core of Indian Polity for SSC CGL!'